

1. Administrative & Study Identification

Brief Title: Study to Evaluate Efficacy and Safety of Inclisiran in Children With Heterozygous Familial Hypercholesterolemia **Full Study Title:** Two Part (Double-blind Inclisiran Versus Placebo [Year 1] Followed by Open-label Inclisiran [Year 2]) Randomized Multicenter Study to Evaluate Safety, Tolerability and Efficacy of Inclisiran in Children (6 to Less Than 12 Years) With Heterozygous Familial Hypercholesterolemia and Elevated LDL-Cholesterol **Short Title/Acronym:** ORION-20 **Protocol Number:** CKJX839C12303 **Posting ID:** 634562336950509525 **NCT Number:** NCT06597019 **Trial Status:** RECRUITING **Sponsor/Company:** Novartis **Investigational Product:** Inclisiran (KJX839) **Trade Name:** Leqvio **ATC Code:** C10AX16 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/leqvio-epar-product-information_en.pdf **Phase of Development:** Phase III **Medical Monitor:** Novartis Study Contact

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of inclisiran compared to placebo on reducing low-density lipoprotein cholesterol (LDL-C) measured by percentage change from baseline to Day 330. **Secondary Objectives:** To evaluate the safety, tolerability, and efficacy of inclisiran, including time-adjusted percent change in LDL-C, absolute change in LDL-C, and percent change in PCSK9.

2.2 Background & Rationale Heterozygous Familial Hypercholesterolemia (HeFH) is an inherited genetic disorder caused by inheriting one abnormal copy of a gene, most commonly the LDLR gene, responsible for clearing low-density lipoprotein (LDL) cholesterol from the blood. This genetic defect leads to a lifelong elevation of plasma LDL cholesterol levels, significantly increasing the risk of premature atherosclerotic cardiovascular disease, including heart attacks and strokes, often in early to mid-adulthood. This study aims to provide clinical evidence for the safety, tolerability, and efficacy of inclisiran (a PCSK9-targeted siRNA) in pediatric patients aged 6 to <12 years with HeFH who remain with elevated LDL-C despite receiving standard of care background lipid-lowering therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (Year 1), Single-group (Year 2) **Blinding:**

Double-blind (Participant, Investigator) during Year 1; Open-label during Year 2 **Randomization:** Randomized (Parallel Assignment in Year 1)

3.2 Planned Enrollment & Duration Trial Status: RECRUITING **Target N:** 51 participants **Number of Sites:** Multicenter (Europe, US, Rest of World) **Estimated Study Start:** 09-Dec-2024 **Estimated Primary Completion:** 15-Apr-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants, 6 to <12 years of age at screening. 2. HeFH diagnosed either by genetic testing or on phenotypic criteria. 3. Fasting LDL-C >130 mg/dL (3.4 mmol/L) at screening. 4. For participants 8 to <12 years: on an optimal dose of statin (investigator's discretion) unless statin intolerant, with or without other lipid-lowering therapy (e.g., ezetimibe). For participants <8 years, the use of background lipid-lowering treatment is based on investigator's discretion. 5. Participants on lipid-lowering therapies (such as statin and/or e.g. ezetimibe) must be on a stable dose for $\geq$ 30 days before screening with no planned medication or dose changes during study participation.

4.2 Exclusion Criteria 1. Previous treatment (within 90 days of screening) with monoclonal antibodies directed towards PCSK9. 2. Secondary hypercholesterolemia (e.g., hypothyroidism or nephrotic syndrome). 3. Homozygous familial hypercholesterolemia (HoFH). 4. Body weight <16 kg at the screening and/or randomization (Day 1) visit. 5. Active liver disease defined as any known current infectious, neoplastic, or metabolic pathology of the liver or unexplained alanine aminotransferase (ALT), aspartate aminotransferase (AST) elevation >3x ULN, or total bilirubin elevation >2x ULN (except patients with Gilbert's syndrome). 6. Pregnant or nursing females. 7. Recent and/or planned use of other investigational medicinal products or devices.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Inclisiran sodium 300 mg for participants with body weight $\geq$ 23 kg, and 180 mg for participants with body weight <23 kg. Administered on Days 1, 90, and 270 during Year 1 (Double-blind). In Year 2 (Open-label), inclisiran is given at Days 450 and 630. (Note: The placebo arm transitions to inclisiran on Days 360, 450, and 630). **Route of Administration:** Subcutaneous (s.c.) injection. **Duration of Treatment:** 2 years (1 year double-blind followed by 1 year open-label).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care background lipid-lowering therapy (e.g., statins, ezetimibe) must

be maintained at a stable dose. **Prohibited:** Use of PCSK9-directed monoclonal antibodies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -30 to 0	Informed Consent/ Assent, Medical History, Fasting LDL-C Labs
Baseline	Day 1	Randomization, First Dose (Inclisiran or Placebo)
Interim 1 (Year 1)	Days 90 & 270	Double-blind Study Drug Administration, Safety & Efficacy Labs
Primary Endpoint	Day 330	Fasting Lipid Panel (LDL-C measurement for Primary Endpoint)
Interim 2 (Year 2)	Days 360, 450, 630	Transition to Open-label; Open-label Study Drug Administration
End of Study	Up to Day 720	Final Safety Labs, Efficacy Assessment, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Percentage change in LDL-C from baseline to Day 330 (Year 1): Demonstrate superiority of inclisiran compared to placebo in reducing LDL-C [percent change] at Day 330.

7.2 Secondary & Exploratory Endpoints 1. Time-adjusted percent change in LDL-C from baseline after Day 90 and up to Day 330 (Year 1). 2. Absolute change in LDL-C from baseline to Day 330 (Year 1). 3. Percent change in PCSK9 from baseline to Day 330 (Year 1). 4. Percent change in total cholesterol, non-HDL-C, triglycerides, HDL-C, and VLDL-C from baseline to each assessment time up to Day 720 (Year 2).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of general safety and tolerability, including injection site reactions and lab abnormalities. **Serious Adverse Events (SAEs):** Standard global reporting timelines (typically 24 hours of site awareness). **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to oversee pediatric patient safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=51$ is calculated to provide adequate power to detect the superiority of inclisiran over placebo in lowering LDL-C. **Handling of Missing Data:** Standard handling utilizing Mixed Model

for Repeated Measures (MMRM) or Multiple Imputation for missing continuous lipid data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard periodic onsite and remote monitoring visits to ensure protocol compliance and data integrity. **Audit**

Trail: Validated eCRF completion, automated edit checks, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: CKJX839C12303 **Posting ID:** 634562336950509525 **Registry Number:** NCT06597019 **Sponsor/Lead Organization:** Novartis **Brief Title:** Study to Evaluate Efficacy and Safety of Inclisiran in Children With Heterozygous Familial Hypercholesterolemia **Official Regulatory Title:** Two Part (Double-blind Inclisiran Versus Placebo [Year 1] Followed by Open-label Inclisiran [Year 2]) Randomized Multicenter Study to Evaluate Safety, Tolerability and Efficacy of Inclisiran in Children (6 to Less Than 12 Years) With Heterozygous Familial Hypercholesterolemia and Elevated LDL-Cholesterol

12. Clinical Framework (Familial Hypercholesterolemia Specific)

Primary Condition: Familial Hypercholesterolemia - Heterozygous **Preferred UMLS Name:** Heterozygous familial hypercholesterolemia

SNOMED ID: 238079002 **MeSH Code:** C06.552.417 **Disease Definition:** Heterozygous Familial Hypercholesterolemia (HeFH) is an inherited genetic disorder caused by inheriting one abnormal copy of a gene, most commonly the LDLR gene, responsible for clearing low-density lipoprotein (LDL) cholesterol from the blood. This genetic defect leads to a lifelong elevation of plasma LDL cholesterol levels, significantly increasing the risk of premature atherosclerotic cardiovascular disease, including heart attacks and strokes, often in early to mid-adulthood. **Diagnosis**

Threshold: Fasting LDL-C >130 mg/dL (3.4 mmol/L) **Medical Methodology:** Subcutaneous siRNA targeted therapy inhibiting PCSK9 production **Optimization Target:** Maximal reduction in LDL-C on top of stable GDMT (statins +/- ezetimibe)

13. Study Procedures & Methodology

Management Pathway: Standard lipid-lowering clinical pathway combined with PCSK9 inhibition. **Education Protocol (HCP):** Site

initiation visits covering study drug preparation (weight-based dosing), administration, and pediatric considerations. **Education Protocol (Patient):** Age-appropriate assent and parent/guardian informed consent; instructions for fasting prior to lab visits. **Quality Audit Mechanism:** Routine sponsor oversight, centralized data checks, and drug accountability tracking.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 720 days (2 Years) **Onsite Visit Cadence:** Days 1, 90, 270, 330, 360, 450, 630, and final completion **Remote Monitoring Frequency:** Intermittent monitoring/check-ins between dosing schedules per protocol. **Checklist Review Interval:** Regular review of concomitant medications and adverse events at every visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				